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Revealing Choice Bracketing†

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Experiments suggest that people fail to take into account interdependencies between their choices—they do not broadly bracket. Researchers often instead assume people narrowly bracket, but existing designs do not test it. We design a novel experiment and revealed preference tests for how someone brackets their choices. In portfolio allocation under risk, social allocation, and induced-value shopping experiments, 40–43 percent of subjects are consistent with narrow bracketing, and 0–16 percent with broad bracketing. Adjusting for each model's predictive precision, 74 percent of subjects are best described by narrow bracketing, 13 percent by broad bracketing, and 6 percent by intermediate cases. (*JEL* D12, D81, D91)

Individuals face many interconnected decisions. How an individual takes into account the interdependencies when choosing, or how they *bracket* these choices, significantly influences their decision-making process. Bracketing determines which outcomes are evaluated as gains or losses and fair or unfair and also plays a role in measuring parameters like risk aversion. Nearly every behavioral model and most "rational" ones require some assumption about how people bracket choices.

There are many ways to bracket. Optimal decision-making requires that people *broadly bracket*, considering every feasible combination of choices and selecting the best. The most common alternative is that people *narrowly bracket* by making each decision without considering any interdependencies. However, these two extremes are far from exhaustive. For instance, Barberis, Huang, and Thaler (2006) and Rabin and Weizsäcker (2009) propose a hybrid of the two called *partial-narrow bracketing*.

Most experimental evidence interpreted as being *for* narrow bracketing is actually evidence *against* broad bracketing. This evidence, surveyed in Section II, comes

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mainly from studies that follow a similar design to Tversky and Kahneman (1981, Problem 3) or Kahneman and Tversky (1979, Problems 11–12). In the former, each subject makes two concurrent choices, and one pair of choices generates a distribution over outcomes dominated by another pair. They find that many subjects choose the dominated pair. In the latter, two groups of subjects face choices between lotteries that are economically identical but differ in how payments are divided between an endowed income and an active choice. The two groups make different choices. Both designs provide evidence against broad bracketing. However, narrow bracketing makes no testable predictions in either design: any choices are consistent with it. This leaves open the question of whether narrow bracketing is a good description of behavior.

We propose a theoretical framework and experimental design to test how an individual brackets. Every subject makes several decisions, each comprised of one or more parts. Their payoff from a decision is determined by the sum of the items chosen in its parts. A subject is consistent with narrow bracketing if they maximize a preference relation in each part, and consistent with broad bracketing if they optimize after integrating all parts of the decision into a single feasible set. We characterize all the testable predictions for narrow, broad, and partial-narrow bracketing. For example, we check for narrow bracketing by performing a standard revealed preference test on a dataset of their choices that treats each part as an independent observation. The results provide individual-level, nonparametric tests that rely only on monotonicity of the underlying preferences. They provide the theoretical basis for an experimental design that tests all three models of choice bracketing.

To implement these tests, we conduct experiments in which subjects make five decisions consisting of one or two parts. Every part is a budget set, and integrating the choices only requires addition. Unlike the existing designs described above, narrow and broad bracketing make distinct predictions in each of the two-part decisions. Every good in the first part is a perfect substitute for a good in the second. Broad bracketers recognize and take full advantage of this substitutability, while narrow bracketers do not.

We apply our design and tests to portfolio choices among risky assets (Choi et al. 2007), dividing money between two anonymous other subjects (Andreoni and Miller 2002; Fisman, Kariv, and Markovits 2007), and standard consumer problems with induced values over bundles. The last allows us to conduct even more powerful tests of bracketing because preferences are known. Across the three experiments, 40–43 percent of subjects are consistent with narrow bracketing, 0–16 percent are consistent with broad, 1–32 percent are consistent with partial-narrow but neither broad nor narrow, and 24–44 percent are consistent with none of the three. No subjects are consistent with *both* broad and narrow bracketing. We then determine which model best describes each subject's bracketing based on both the number of errors they make relative to each model and the predictive precision of that model (Selten 1991). We classify 6–13 percent of subjects to broad bracketing, 68–78 percent to narrow bracketing, and only 3–9 percent to partial-narrow bracketing, with variation across experiments.[1]

[1] The remaining subjects are poorly described by all models and left unclassified.

Failing to bracket broadly has consequences for welfare. Economists as early as Smith (1776) argue that agents benefit from specialization. A broad bracketer specializes by purchasing as much of a good as possible when it is relatively cheap before purchasing any when it has a higher opportunity cost. This increases utility in the same way that comparative advantage increases total productivity in classic trade models.[2] Broad bracketers apply Smith's insights about the productivity increases from specialization to individual decision-making, while narrow bracketers fail to reap the gains from specialization.[3] Our approach allows us to quantify the magnitude of these losses. For instance, the subjects classified as narrow bracketers made an average of $1.29 less than the broad bracketers from the two-part decisions in the Shopping Experiment, more than 10 percent of the variable payment.

Because our tests are at the individual level, they can provide evidence on why so many people bracket narrowly despite the gains from broad bracketing. In online follow-up experiments, we recorded choice process data and implemented a nudge to encourage subjects to "examine both parts" of the decision. While the nudge increased the proportion of subjects who looked at both parts, it had a limited effect on the rate of broad bracketing. In one arm of the online Risk Experiment, we observed that about a quarter of the narrow bracketers had enough information to bracket broadly but did not do so. In the online Shopping Experiment, a similar fraction used a calculator to compute the payoff of one or more bundles that were only feasible at the decision level and not in any part on their own. Both these behaviors indicate some consideration of how choices from both parts combine to determine payoffs. This suggests that not all narrow bracketing results from simply looking at one part of a decision at a time and making an optimal choice for that part on its own. Instead, it suggests that some individuals consider both decision problems, yet ultimately implement choices that are only compatible with narrow bracketing.

Bracketing also affects how one interprets behavior. A narrow bracketer's reluctance to take a small yet actuarially favorable gamble indicates only slight risk aversion. However, for broad bracketers, the rejection of the same gamble signifies extreme risk aversion, given its minimal impact on their overall wealth (Rabin 2000). Our approach enables us to separate underlying preferences from bracketing, with implications for measuring risk or inequity aversion.

This separation allows us to measure heterogeneity in bracketing directly and to account for it when inferring preferences. To illustrate the importance of doing so, consider the Social Experiment. In part 2 of Decision 1 (see Table 2), subjects on average allocate unequally between the two anonymous others, dividing $16 into $6.80 and $9.20. If all subjects bracketed narrowly, this would suggest a lack of concern for equity. However, we classify 10 percent of subjects as broad bracketers and 75 percent as narrow bracketers. The narrow bracketers on average allocate $7.75 and $8.25 to each person, and the broad bracketers all achieve perfectly equal allocations in the decision by allocating a $14–$2 split in that part and a $0–$12 split in the other. After taking bracketing into account, all these subjects are consistent with inequity aversion. The only existing work that estimates heterogeneity in bracketing, Rabin

[2] Baron and Kemp (2004) provide a survey measure of understanding of specialization and show it is correlated with attitudes to free trade.

[3] We thank a referee for pointing to this analogy.

and Weizsäcker (2009), estimates a between-subjects, parametric, structural model and assumes that all subjects have the same preferences. In contrast, our approach is individual level, nonparametric, and allows heterogeneity in preferences.

Most applications of prospect theory explicitly assume narrow bracketing (Camerer 2004, Table 5.1; O'Donoghue and Sprenger 2018, Sections 3.4 and 4.5). Since it is impossible to avoid all other risks, a broad bracketer will not exhibit noticeable small-stakes loss aversion (Barberis, Huang, and Thaler 2006). At the same time, other applications of prospect theory, such as casino gambling (Barberis 2012; Ebert and Strack 2015), require broader bracketing. Our results suggesting heterogeneity in bracketing may reconcile the results rejecting broad bracketing (e.g., Tversky and Kahneman 1981) and those more supportive of it (e.g., Heimer et al. 2020 and Baillon, Halevy, and Li 2022).

Choice bracketing is also important for social decisions. For example, Sobel (2005, p. 400) remarks that in experiments studying social decisions, a subject should "maximize her monetary payoff in the laboratory and then redistribute her earnings to deserving people later … if the concern for inequity was 'broadly bracketed."' However, the literature he surveys and our analysis both find evidence consistent with subjects caring about narrowly bracketed equity (e.g., Charness and Rabin 2002; Bolton and Ockenfels 2000).

Standard economic analyses make assumptions about bracketing as well. As noted by a recent behavioral economics textbook, "Although choice bracketing has been largely ignored in economics, models in economic theory often implicitly assume and invoke it" (Dhami 2016, p. 1469). Many analyses exclude some related choices from the model yet interpret results in terms of a fully rational choice. Our results suggesting that most people are narrow bracketers lend support to this approach, with the caveat that some seem best modeled as broad bracketers.

I. Testing Choice Bracketing

We consider a decision-maker (DM) who faces $T \geq 1$ decision problems involving alternatives contained in $\mathbb{R}^n_+$. Decision t consists of $K_t \geq 1$ parts. Formally, the feasible set of part k from decision t is $B^{t,k}$, assumed to be compact and nonempty. The DM chooses one alternative, denoted $x^{t,k}$, from each part, $B^{t,k}$. Thus, the analyst observes the *dataset*

$$\mathcal{D} = \left\{\left(x^{t,k}, B^{t,k}\right)\right\}_{(t,k)}$$

indexed by decisions and parts and where $x^{t,k} \in B^{t,k}$. They face parts concurrently, and their payoff in decision t depends only on the sum of their choices across parts of the decision,

$$x^t = \sum_{k=1}^{K_t} x^{t,k},$$

which we call the "final alternative." A real-world analog consists of a scanner dataset with purchases at different stores (parts, indexed by k) in a given time period (a decision, indexed by t), and the shopper consumes these purchases after buying at all stores but before the next period.

A DM *broadly brackets* if they maximize an increasing utility function over the set of feasible final alternatives for each decision problem,

$$B^t = \left\{ \sum_{k=1}^{K_t} y^{t,k} : y^{t,k} \in B^{t,k} \right\}.$$

In contrast, they *narrowly bracket* if they choose the alternative in part k of decision t that maximizes an increasing utility function over $B^{t,k}$.[4] Both require that the DM is "rational" in that they maximize some well-behaved preference relation, but neither requires any further assumptions about preferences beyond monotonicity. A narrow bracketer optimizes part by part, while a broad bracketer does so decision by decision.

Our goal is to test which, if any, models of bracketing can explain a subject's choices. Formally, a dataset $\mathcal{D}$ is *rationalized by broad bracketing* if there exists an increasing utility function $u : \mathbb{R}^n_+ \rightarrow \mathbb{R}$ so that

$$x^t = \arg\max_{x \in B^t} u(x)$$

for every decision t, and a dataset $\mathcal{D}$ is *rationalized by narrow bracketing* if there exists an increasing $u : \mathbb{R}^n_+ \rightarrow \mathbb{R}$ so that

$$x^{t,k} = \arg\max_{x \in B^{t,k}} u(x)$$

for every part k and decision t.[5] The next two subsections provide necessary and sufficient conditions for $\mathcal{D}$ to be rationalized by either of the two.

A. *Predictions*

We first provide necessary conditions for rationalization by broad and narrow bracketing. The first two predictions build on the observation that the DM's choice reveals their preference among the set of alternatives over which they optimize. The predictions combine the weak axiom of revealed preference (WARP) with a form of bracketing.

Prediction 1 (NB-WARP)**:** Suppose that $\mathcal{D}$ is rationalized by narrow bracketing. If $\left(x^{t,k}, B^{t,k}\right), \left(x^{t',k'}, B^{t',k'}\right) \in \mathcal{D}$ and $x^{t',k'} \in B^{t,k} \subseteq B^{t',k'}$, then $x^{t,k} = x^{t',k'}$.

Prediction 2 (BB-WARP)**:** Suppose that $\mathcal{D}$ is rationalized by broad bracketing. For decisions t, t' in $\mathcal{D}$, if $x^{t'} \in B^t \subseteq B^{t'}$, then $x^t = x^{t'}$.

[4] A narrow bracketer acts as if they perceive alternatives correctly and maximizes a well-behaved preference over them, but misperceives the budget set. In contrast, a DM who misperceived correlation, e.g., Eyster and Weizsäcker (2016) or Ellis and Piccione (2017), perceives the choice set correctly but misperceives the alternatives themselves.

[5] These definitions assume that each choice represents strict preference, but one can easily generalize them and the results that follow to allow for indifference.

Narrow bracketing requires that WARP holds when comparing any pair of parts of decisions, even when they belong to economically different decisions. Broad bracketing implies that WARP holds at the decision level, comparing final alternatives that are feasible in both aggregate budget sets.

The next prediction reflects the appropriate manifestation of monotonicity in our setting.

Prediction 3 (BB-Mon)**:** Suppose that $\mathcal{D}$ is rationalized by broad bracketing. For any decision t in $\mathcal{D}$ and any $y \in B^t$, if $y \geq x^t$, then $y = x^t$.

BB-Mon requires that the subject chooses on the frontier of their aggregate budget set in a given decision.[6]

B. *Characterization*

The above predictions are necessary but not sufficient conditions for each type of bracketing. We obtain tight characterizations by extending the logic of the above to include indirect implications, in the same manner that strong axiom of revealed preference (SARP) extends WARP. Theorem 1 shows that these are necessary and sufficient conditions for a given dataset to be rationalized by a particular form of bracketing.

Our tests are based on applying SARP to an ancillary dataset. Say that a bundle x is *directly revealed preferred to* y in a dataset $\mathcal{D}'$ consisting of single-part decisions, written $x P^{\mathcal{D}'} y$, if either $x \geq y$ and $x \neq y$ or there exists $(x,B) \in \mathcal{D}'$ so that $y \in B \backslash \{x\}$. A dataset $\mathcal{D}'$ *satisfies SARP* if the binary relation $P^{\mathcal{D}'}$ is acyclic.

We say that $\mathcal{D}$ satisfies BB-SARP if the ancillary dataset

$$\mathcal{D}^{BB} = \left\{\left(x^t, B^t\right)\right\}_{t \in T}$$

satisfies SARP. Similarly, we say that $\mathcal{D}$ satisfies NB-SARP if SARP is satisfied by the ancillary dataset

$$\mathcal{D}^{NB} = \left\{\left(x^s, B^s\right)\right\}_{s=1}^{\sum_{t=1}^{T} K_t},$$

where for each part (t,k), there exists a unique s so that $\left(x^s, B^s\right) = \left(x^{t,k}, B^{t,k}\right)$. In $\mathcal{D}^{NB}$, each part of every decision is treated as a separate, independent observation, whereas each decision is in $\mathcal{D}^{BB}$.

THEOREM 1: *The following are true:*

(*i*) *The dataset $\mathcal{D}$ satisfies BB-SARP if and only if $\mathcal{D}$ is rationalizable by broad bracketing*, *and*

[6] Narrow bracketing predicts a similar condition at the part level. Our experimental implementation forces it to hold, so we do not formally include it here.

(*ii*) *The dataset $\mathcal{D}$ satisfies NB-SARP if and only if $\mathcal{D}$ is rationalizable by narrow bracketing.*

Theorem 1 characterizes the complete testable implications of broad and narrow bracketing. Moreover, both conditions are readily applied using standard computational tools.[7] One can make tighter predictions by imposing more structure on the utility function that rationalizes the data. For instance, we impose that preferences are symmetric when we apply these tests to our experiments. This enables stronger versions of the two tests.[8]

C. *Partial-Narrow Bracketing*

We also consider an intermediate model of bracketing lying between the two extremes, called *α-partial-narrow bracketing* (*PNB*). Inspired by Barberis, Huang, and Thaler (2006); Barberis and Huang (2009); and Rabin and Weizsäcker (2009), the dataset $\mathcal{D}$ is *rationalized by α-partial-narrow bracketing* if

$$(1) \qquad \left(x^{t,1}, \ldots, x^{t,K_t}\right) = \underset{y^{t,k} \in B^{t,k} \, \forall k}{\arg\max} \, \alpha \sum_{k=1}^{K^t} u\left(y^{t,k}\right) + (1-\alpha) u\left(\sum_{k=1}^{K^t} y^{t,k}\right)$$

for all t.[9] A PNB DM's choices maximize a weighted average of the utility of the final alternative and of the utility of their choice in each part.

A broad bracketer takes into account that good i in part k is a perfect substitute for good i in part k', whereas a narrow bracketer does not consider any complementarities across parts of the decision.[10] In contrast, a PNB DM views the same good in different parts as imperfect substitutes for each other. They act as if they make a single choice in decision t of an element of $\mathbb{R}_+^{nK_t}$, but they maximize a utility function derived from u rather than u itself. In contrast, a broad bracketer acts as if they choose a single element of $\mathbb{R}_+^n$, and a narrow bracketer as if they make K_t independent choices of elements of $\mathbb{R}_+^n$.[11]

An algorithm *tests* whether $\mathcal{D}$ is rationalizable by α-partial-narrow bracketing if it inputs a dataset $\mathcal{D}$ and (deterministically) outputs one when $\mathcal{D}$ is rationalizable by α-partial-narrow bracketing and outputs zero otherwise.

[7] We note that B^t typically has a piecewise linear budget line even if each $B^{t,k}$ is a standard budget set.

[8] Specifically in $\mathbb{R}_+^2$, if $(x,y) \, P^{\mathcal{D}}(x',y')$, then $(y,x) \, P^{\mathcal{D}}(x',y')$, $(x,y) \, P^{\mathcal{D}}(y',x')$, and $(y,x) \, P^{\mathcal{D}}(y',x')$. In online Appendix B, we derive some immediate implications of symmetry.

[9] The first two papers consider an average of the certainty equivalents, $\left[u^{-1}\left(E\left[u(x^t)\right]\right) + b_0 \sum_{k=1}^{K^t} v^{-1}\left(E\left[v(x^{t,k})\right]\right)\right]$. Unlike our specification, they allow for the narrow utility function to differ from the broad utility function and assume expected utility. The algorithm in Theorem 2 can be adapted to allow for different narrow and broad utility functions ($u \neq v$) at the cost of less predictive power. Our specification is closest to Rabin and Weizsäcker (2009, p. 1513), although they also assume expected utility.

[10] Köszegi and Matějka (2020) also base their approach to mental budgeting on substitutability between parts.

[11] Recent models by Vorjohann (2020) and Zhang (2023) have this feature as well but with different utilities over $\mathbb{R}_+^{nK_t}$ than equation (1). They capture departures from broad bracketing by either substituting in a reference point for other parts or by imposing separability where u is inseparable. In a two-part decision, these models are as follows. Vorjohann (2020) considers a DM who maximizes $v\left(x^{t,1},x^{t,2}\right) = u\left(x^{t,1},r^2\right) + u\left(r^1,x^{t,2}\right)$ for reference points $r^1 \in B^{t,1}$ and $r^2 \in B^{t,2}$, instead of $v^{BB}\left(x^{t,1},x^{t,2}\right) = u\left(x^{t,1},x^{t,2}\right)$. Zhang (2023), for the specification most similar to our setting, considers a DM who maximizes $U\left(x^{t,1},x^{t,2}\right) = E_{x^{t,1}}\left[u\left(z + u^{-1}\left(E_{x^{t,2}}\left[u(y)\right]\right)\right)\right]$ instead of $U^{BB}\left(x^{t,1},x^{t,2}\right) = E_{x^{t,1}+x^{t,2}}\left[u(z)\right]$, where E_x is the expectation operator with respect to the lottery corresponding to x.

THEOREM 2: *There exists an algorithm that tests whether $\mathcal{D}$ is rationalizable by α-partial-narrow bracketing.*

We construct the algorithm in Appendix A. The algorithm reduces the problem of determining whether $\mathcal{D}$ is consistent with α-partial-narrow bracketing to that of a standard linear programming problem. It runs in polynomial time with a fixed set of possible outcomes of each decision and its parts. We elaborate on the ideas behind it and provide a formal proof in Appendix A.

The algorithm takes advantage of the formal similarity between equation (1) and expected utility. Specifically, an $\alpha-$PNB DM with utility u chooses $x^{t,1}$ in part 1 and $x^{t,2}$ in part 2 if and only if an expected utility DM with Bernoulli index u chooses $(\psi\alpha, x^{t,1}; \psi\alpha, x^{t,2}; \psi(1-\alpha), x^t)$ from the menu of lotteries $\{(\psi\alpha, y^1; \psi\alpha, y^2; \psi(1-\alpha), y^1 + y^2) : y^1 \in B^{t,1} \text{ and } y^2 \in B^{t,2}\}$, where $\psi = (1+\alpha)^{-1}$. The algorithm first transforms the original dataset to an ancillary dataset where each decision is a single choice from a menu of lotteries. Each lottery in the menu is structured so that an alternative in part k occurs with probability proportional to α for each part k and their sum occurs with probability proportional to $(1-\alpha)$. The DM's choice is the lottery over their choices in each part and their sum. Existing results, e.g., Clark (1993), provide necessary and sufficient conditions for the ancillary dataset to be rationalizable by expected utility. We show that whether or not it can be rationalized can be determined by examining the solution to a linear program.

We also consider a related intermediate model of bracketing, "PNB-Personal Equilibrium (PNB-PE)." Formally,

$$x^{t,k} = \underset{y^{t,k} \in B^{t,k}}{\arg\max} \left[\alpha u(y^{t,k}) + (1-\alpha) u\left(\sum_{k' \in \{1 \le k' \le K^t : k' \neq k\}} x^{t,k'} + y^{t,k}\right)\right]$$

for every part k of decision t. While the functional form is similar to PNB, the DM acts as if they make K_t independent choices of elements of $\mathbb{R}^n_+$, as a narrow bracketer would. Their choices are the personal equilibrium (PE) of an intrapersonal game (Köszegi and Rabin 2006; Lian 2019). A PNB-PE DM only considers the utility effect of $x^{t,-k}$ when choosing $x^{t,k}$, ignoring that they could change $x^{t,-k}$ in response to their choice. A similar algorithm to the one that tests α-PNB can test α-PNB-PE.

II. Existing Evidence on Bracketing

The models we consider capture different forms of what Read, Loewenstein, and Rabin (1999) call choice bracketing: how a person groups "individual choices together into sets." We first discuss three strands of research into static, concurrent decisions to which our setting directly applies before turning to the important case of dynamic, or temporal, bracketing.

Most existing direct evidence of nonbroad choice bracketing is derived from Tversky and Kahneman (1981), in which subjects face the decision problem described in Table 1.

Table 1—Tversky and Kahneman's Decision Problem

Imagine that you face the following pair of concurrent decisions. First, examine both decisions, then indicate the options you prefer.	
Decision (i). Choose between:	
A. A sure gain of \$240 [85 percent]	B. 25 percent chance to gain \$1,000 and 75 percent chance to gain nothing [16 percent]
Decision (ii). Choose between:	
C. A sure loss of \$750 [13 percent]	D. 75 percent chance to lose \$1,000 and 25 percent chance to lose nothing [87 percent]

This design fits into our theoretical setting as a single-decision dataset with $B^{1,1} = \{A,B\}$ and $B^{1,2} = \{C,D\}$. The choice combination (A,D) made by 73 percent of their subjects generates a first-order stochastically dominated distribution over outcomes compared to the pair of choices (B,C) and so violates BB-Mon and BB-SARP. Their design cannot falsify narrow bracketing since every combination of choices satisfies NB-SARP. Without further restrictions, their results are uninformative about the choice bracketing of subjects who do not choose A and D. Yet about 70 percent of subjects made non-(A,D) choices in incentivized follow-up experiments (Rabin and Weizsäcker 2009; Koch and Nafziger 2019).

A related set of experiments considers how an exogenously fixed quantity, like a monetary endowment, an asset, or a background risk, affects a person's choice when separated from the description of available alternatives.[12] For instance, studies suggest subjects behave as if they do not incorporate their endowment in risk-taking choices (Kahneman and Tversky 1979, Problems 11–12), in social allocation tasks (Exley and Kessler 2018), and in labor supply choices (Fallucchi and Kaufmann 2021). These designs fit into our theoretical framework as decisions in which one part is a singleton set (the endowment) and one part a nonsingleton set (the active choice), and these papers provide evidence against broad bracketing.

The final body of evidence shows how failures of fungibility affect choice (Thaler 1999). Studies of the "flypaper effect" suggest that transfers earmarked for a particular type of spending tend to actually be spent there (Hines and Thaler 1995; Abeler and Marklein 2017). Empirical evidence from consumption choices after unexpected price changes support the lack of fungibility (Hastings and Shapiro 2013, 2018).[13]

This evidence has a more complicated relationship with our setting. Thaler (1985; 1999) and others (Galperti 2019; Köszegi and Matějka 2020) explain the evidence through mental accounting (or budgeting). At its most general, this refers to a person breaking up the overall decision into categories. In contrast, narrow choice bracketing is a failure to aggregate smaller parts into a larger decision. These are not mutually exclusive. If categories and parts coincide, our model of narrow choice bracketing provides an extreme model of mental accounting. But in general, mental accounting is consistent with either broadly bracketing across parts

[12] Related theoretical results by Samuelson (1963); Rabin (2000); Safra and Segal (2008); and Mu et al. (2020) suggest that behavior in small-stakes bets cannot be reconciled with behavior in large-stakes bets and broad bracketing. The results of Gneezy and Potters (1997) and Benartzi and Thaler (1995) provide support for their theoretical predictions.

[13] Evers, Imas, and Kang (2022) suggest that similarity mediates how people group outcomes into categories.

(Thaler 1985; p. 207) or with narrowly bracketing each part (e.g., as described by Corollary 1 of Köszegi and Matějka 2020).[14]

Choice bracketing is also relevant in dynamic settings. Gneezy and Potters (1997) show that subjects make different choices period by period than when they must choose in advance, evidence of nonbroad bracketing.[15] In contrast, recent evidence from Heimer et al. (2020) (building on Barberis 2012; Ebert and Strack 2015) suggests that subjects behave as if they do not narrowly bracket future risk-taking opportunities. In financial decision-making, Shefrin and Statman (1985) and Odean (1998) document the disposition effect, and Thaler and Johnson (1990) and Imas (2016) document the house money effect. Either effect is inconsistent with both fully narrow and fully broad bracketing.

Our approach can be generalized to a dynamic setting, but two key complications arise. First, the setting should comprise decision trees rather than sets of alternatives. Second, behavior may be dynamically inconsistent. Our approach can be adapted but would apply only under dynamic consistency. Future work should incorporate dynamic inconsistency.

We note that there are many ways to fail to bracket broadly in a dynamic setting. For instance, an agent may only look backward and take into account the results of their past decisions but neglect to consider how their current decision interacts with their future ones. Or they may only look forward and take into account future interactions but not past results. Even more combinations are possible: they could look neither forward nor backward or only look forward a certain number of periods.

III. Experimental Design

We design and conduct three experiments to test the models of bracketing in different domains of choice. In each experiment, a participant faces five decision rounds, each consisting of one or two parts.[16] Each part consists of all feasible integer-valued bundles of two goods obtained from a linear (or in one case, a piecewise linear) budget set. At the end of the experiment, exactly one round is randomly selected for payment, which we call the "round that counts." We sum all goods purchased in all parts of the decision in the round that counts to obtain the final bundle that determines payments.[17] By design, there are no complementarities across decisions. We implement this experimental design to study choice bracketing in three domains of interest: portfolio choice under risk (Risk), a social allocation

[14] One can extend our framework by replacing the "SARP" part of NB- or BB-SARP with a revealed preference test for mental accounting, such as that of Blow and Crawford (2018). Then, we have two joint tests: one for mental accounting where the budgets for each category are determined part by part, and one for mental accounting where the budgets for each category are determined at the decision level.

[15] The comparison in Cox, Sadiraj, and Schmidt (2015) of their "One Task" treatment and their "Pay all sequentially" treatment provides between-subjects evidence that fails to reject the null that every subject narrowly brackets. They note that this is consistent with prior literature that finds little evidence for "wealth effects" (i.e., violations of narrow bracketing) in economics experiments that pay all choices sequentially.

[16] Compared to choice-from-budget-set experiments like Choi et al. (2007), each subject faces fewer rounds of decisions in our experiment, and each budget set is coarser to allow us to conduct the experiment on paper. We made this design choice to minimize potential for learning to bracket and also to slow down subjects' progression through the experiment to encourage slow and deliberate decision-making.

[17] In the Social Experiment we modify this procedure slightly: one person-decision pair per anonymous group of two subjects is randomly selected to determine payments of another anonymous group of two subjects.

task (Social), and a consumer choice experiment in which we induced subjects' values (Shopping).

In the Risk Experiment, each part of every decision asks the subject to choose an integer allocation of tokens between two assets. Each asset pays off on only one of two equally likely states: Asset A (or C) pays out only on a die roll of 1–3, whereas Asset B (or D) pays out only on a die roll of 4–6. The payoff of each asset varies across decision problems and across parts. Because each decision problem uses assets with two equally likely states, preferences over portfolios of monetary payoffs for each state should be symmetric across states.

In the Social Experiment, each part of every decision asks the subject to choose an integer allocation of tokens between two anonymous other subjects, Person A and Person B. The value of each token to A and B varies across decision problems and across parts. Because the two recipients are anonymous, we expect preferences to be symmetric across money allocated to A versus B.

In the Shopping Experiment, each part of every decision asks the subject to choose a bundle of integer quantities of fictitious "apples" and "oranges" subject to a budget constraint. The monetary payment for the experiment is calculated from the final bundle in the round that counts according to the function pay $= \frac{2}{5}\left(\sqrt{\#apples} + \sqrt{\#oranges}\right)^2$ dollars.[18] This function induces payoffs that are symmetric in apples and oranges and that are strictly variety seeking. Any subject who prefers more money to less will wish to maximize this payoff function regardless of their underlying utility function.

Our experimental budget sets, summarized in Table 2, allow us to conduct our revealed preference tests in the Risk and Social Experiments and to conduct analogous tests that make use of the induced-value function in our Shopping Experiment. Throughout, we refer to part *k* of decision *t* as Dt.k, or Dt if a round has only one part. The order of decisions and parts was varied across sessions (Table 15 in online Appendix H).

We implemented our experiments on paper and followed up with some robustness treatments online (discussed further in Section V and online Appendix I). In each session, paper instructions were provided and read aloud, subjects were given the opportunity to ask questions privately, and then participants completed a brief comprehension quiz and the experimenter individually checked answers and explained any errors. The comprehension quiz had each subject calculate how payment would be determined from their allocations when a decision involves two parts. This was designed to ensure that each subject was aware of how to calculate payments in these cases but without instructing them to consider all possible combinations of allocations across parts.

Each decision had a cover page indicating the number of parts in the decision, with each part stapled beneath as a separate page. Thus, a subject was always informed when a decision contained multiple parts but could choose whether or not to look at both parts before making allocations. A subject indicated each allocation by highlighting the line corresponding to their choice for each part using a provided

[18] Subjects were provided with a payoff table (Figure 18 in online Appendix H) to calculate the earnings that would result from any possible final bundle, so they could maximize earnings without having to manually compute this function.

Table 2—Experimental Tasks

		Risk			Social			Shopping		
Decision	Part	I	Asset A/C	Asset B/D	I	V_A	V_B	I	p_a	p_o
D1	1	10	$(\$1, \$0)$	$(\$0, \$1.20)$	10	\$1	\$1.20	8	2	1
	2	16	$(\$1, \$0)$	$(\$0, \$1)$	16	\$1	\$1	24	2	2
D2	1	14	$(\$2, \$0)$	$(\$0, \$2)$	14	\$2	\$2	32	2	1 (for first 8), 2
D3	1	10	$(\$1, \$0)$	$(\$0, \$1)$	10	\$1	\$1	30	3	3
	2	10	$(\$1, \$0)$	$(\$0, \$1.20)$	10	\$1	\$1.20	24	3	2
D4	1	16	$(\$1, \$0)$	$(\$0, \$1)$	16	\$1	\$1	12	1	1
D5	1	10	$(\$1, \$0)$	$(\$0, \$1.20)$	10	\$1	\$1.20	48	6	4

I: income for a part (in tokens)

$(\$x, \$y)$ indicates one unit of asset pays $\$x/\$y$ if the die roll is 1–3/4–6

V_A: value/token to A
V_B: value/token to B

p_a: price/apple
p_o: price/orange

highlighter. Subjects were allowed no other aids at their desk when making choices. Only one decision was handed out to a subject at a time, and that decision was collected before the next round was handed out. The order of decisions, and of parts within each decision, was varied across sessions to allow us to test and control for order effects.

Sessions took place in Toronto Experimental Economics Lab and SFU Experimental Economics Lab from June 2019 to February 2020, each taking place in a one-hour block. Subjects were recruited from the labs' student participant pools to participate in one of the three experiments. Instructions, experimental materials, and details of the experimental procedure are provided in online Appendix H.[19]

IV. Results

This section reports the results of our experimental tests of the models of bracketing. For each test, we also compute results allowing for one or two "errors" relative to its requirements. We define an error as how far we would need to move a subject's allocations for them to pass that test, measured in lines on the decision sheet(s).[20] For instance, in Risk and Social, a subject is within one error of passing a test if revising the choice by shifting one token from one asset/person to the other in a single part would lead to choices that pass that test. They are within two errors of passing a test if shifting two tokens from one asset/person to the other, either in the

[19] The three experiments were separately preregistered through the Open Science Framework. Our analysis follows our plans with only minor modifications that do not affect the interpretation of our results. See online Appendix G for details.

[20] Example decision sheets are provided in Figures 10, 11, 14, and 17 in online Appendix H.

same part or in different parts, would lead to choices that pass. For predictions that require Walrasian budget sets, we modify the tests to account for discreteness in our experiment.

In Section V, we examine the robustness of the following analysis to two particular concerns. First, we argue our results are robust to changes in the presentation of the decision. Second, broad bracketing requires allocating all the budget to one good in two-part decisions, and earlier work suggests subjects may be loath to do so. We argue there that neither affects our conclusions much.

A. *Risk Experiment: How Subjects Behave*

Consider Decision 1 in the Risk Experiment. Since Asset B in the first part is a perfect substitute for Asset D in the second, comparative advantage requires that the subject first purchases it in the part where it has a lower opportunity cost. Consequently, a broad bracketer necessarily allocates all of their wealth in the first part to Asset B before they allocate any to Asset D in the second (as required by BB-Mon). If preferences are symmetric across the two equally likely states, then broad bracketing further requires that $x_A^{1,1} = 0$. With risk aversion, broad bracketing makes even stronger predictions. The allocation $x_A^{1,1} = 0$ and $x_C^{1,2} = 14$ obtains a risk-free return of \$14, and any other feasible allocation results in a first- or second-order stochastically dominated distribution over returns at the decision level. In contrast, a narrow bracketer facing Decision 1 allocates at least half of their budget to Asset B in part 1. If they are also risk averse, then they allocate exactly half their budget in part 2 to each asset.

To compare how our subjects perform to this benchmark, Figure 1 (panel A) plots the joint distribution of their allocations in D1. The x-coordinate describes their allocation to Asset A in D1.1, and the y-coordinate their allocation to Asset C in D1.2. The above discussion shows that broad bracketers will have allocations on the y-axis, and a risk-averse broad bracketer will select the allocation $\left(x_A^{1,1}, x_C^{1,2}\right) = (0,14)$. Any narrow bracketer will select an allocation with $x_A^{1,1} \leq 5$, and any risk-averse narrow bracketer will select $x_C^{1,2} = 8$. The plot shows that few subjects are close to consistent with broad bracketing, and only a single subject makes an allocation close to the prediction of risk-averse broad bracketing. Narrow bracketing with risk aversion is consistent with the modal allocation of $\left(x_A^{1,1}, x_C^{1,2}\right) = (4,8)$.

In our design, narrow bracketing makes strong predictions across decisions that have a part in common. For instance, D3.2 and D5 both ask subjects to allocate ten tokens between identical assets with identical prices. A narrow bracketer would make the same allocation in each of the two parts. To illustrate, Figure 1 (panel B) plots each subject's allocation in D3.2 against their allocation in D5. The x-coordinate describes their allocation to Asset C in D3.2, and the y-coordinate their allocation to its counterpart, Asset A in D5. A narrow bracketer's choices will fall on the 45-degree line as required by NB-WARP, and the farther the allocations are from that line, the farther the subject is from narrow bracketing. In the plot, we can see that this prediction of narrow bracketing holds exactly for 54 of the 99 subjects. We visually represent all the data underlying our tests in online Appendix B.

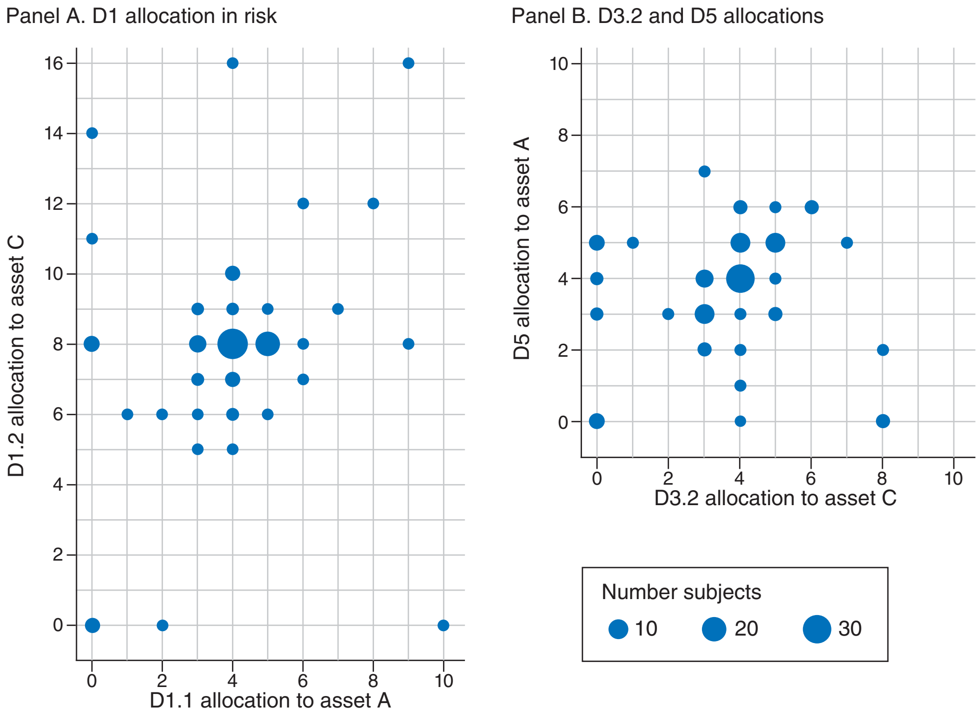

Figure 1. Allocations in Risk

Table 3—Tests of NB-WARP and BB-WARP

	Risk			Social		
Number of errors	0	1	2	0	1	2
NB-WARP (D1.1 and D5)	56	76	89	45	70	77
NB-WARP (D1.2 and D4)	56	74	81	63	78	82
NB-WARP (D3.2 and D5)	54	76	83	49	75	80
NB-WARP (D1.1 and D3.2)	49	76	85	51	83	87
NB-WARP (all)	29	44	61	28	54	64
BB-WARP (D1 and D2)	13	20	87	16	20	94
BB-Mon (D1)	12	13	15	14	14	14
BB-Mon (D3)	14	16	18	17	17	18
BB-Mon (both)	7	8	10	12	12	12
Number of subjects		99			102	

Note: Entries count the number of subjects who pass test at the listed error allowance.

B. *Risk and Social Experiments: Revealed Preference Tests*

We begin by performing the direct revealed preference tests of bracketing developed in Section I: NB-WARP, BB-WARP, and BB-Mon (Table 3).

Very few subjects are consistent with rationality and broad bracketing. There is only a single pair of decisions (D1 and D2) where choices could directly violate

BB-WARP. For that pair, we test BB-WARP by comparing the final bundle for D1 to the final bundle for D2: for any choice of $x^{2,1}$ in D2 with $x_A^{2,1} \leq 8$, the same final bundle can be achieved in D1. In each of Risk and Social, only 20 percent of subjects are within 1 error of passing BB-WARP.[21] Even fewer subjects are consistent with BB-Mon than with BB-WARP. In Risk and Social, respectively, we find that 8 percent and 12 percent of subjects are within 1 error of passing BB-Mon in both decisions. Looking separately at D1 and D3, between 13 and 17 percent of subjects are within 1 error of passing BB-Mon.

All told, the BB-WARP and BB-Mon tests provide evidence showing that 80–92 percent of subjects are not broad bracketers. These rates of violations of broad bracketing are qualitatively similar to, but higher than, those found by Tversky and Kahneman (1981) (73 percent) and Rabin and Weizsäcker (2009) (28–66 percent) and very close to the structural estimates of the latter (89 percent). In these prior experiments, each part consisted of a pairwise choice, so failures to broadly bracket are detected only for a particular range of risk preferences. In contrast, there are many ways a subject could reveal their failure to bracket broadly in our experiments, which gives us more power to detect failures.

While previous work can only falsify broad bracketing, our design allows us to test narrow bracketing as well. We test NB-WARP by comparing the allocations in each of the two parts that appear in multiple decisions. Specifically, NB-WARP requires that a subject makes the same choice in D1.1, D3.2, and D5 and the same choice in D1.2 and D4. Far more subjects pass each NB-WARP test than either BB-WARP or BB-Mon. Between 75 and 77 percent of subjects in Risk and 69 and 81 percent of subjects in Social are within 1 error of passing each of the pairwise NB-WARP tests. Allowing for 1 error, 44 percent and 53 percent of subjects pass all possible NB-WARP tests in Risk and Social, respectively.[22]

Result 1: When allowing for 1 error, 44 percent and 53 percent of subjects are consistent with WARP under narrow bracketing, 20 percent and 20 percent are consistent with WARP under broad bracketing, and 8 percent and 12 percent are consistent with monotonicity under broad bracketing in the Risk and Social Experiments, respectively.

We next conduct our revealed preference tests of the three models considered using the entire set of decisions for each subject. Notice that in both experiments, the alternative (x,y) leads to an identical outcome as the alternative (y,x).[23] As discussed in Section I, we extend the tests so that whenever $(a,b)\,P^{\mathcal{D}}(x,y)$, i.e., (a,b) is directly revealed preferred to (x,y) in the dataset $\mathcal{D}$, we also have $(a,b)\,P^{\mathcal{D}}(y,x)$, as well as $(b,a)\,P^{\mathcal{D}}(x,y)$and $(b,a)\,P^{\mathcal{D}}(y,x)$. This reduces the need to compare across decisions and makes our tests more demanding. For example, all three tests make

[21] Allowing for two errors substantially raises pass rates of BB-WARP in Risk and Social. As predicted by both risk-averse narrow and broad bracketing, many subjects (66 percent in Risk and 84 percent in Social) choose $x_A^{2,1} = 7$. The jump at two errors happens because a subject who allocates $x_A^{2,1} > 8$ trivially satisfies BB-WARP.

[22] In contrast, randomly generated choices have only a 0.4 percent chance of being within one error of passing all NB-WARP tests.

[23] Either leads to a lottery paying x and y with equal probability or to a payment of x to one anonymous individual and y to another.

Table 4—Full Tests of Symmetric Models

	Risk			Social		
Number of errors	0	1	2	0	1	2
NB-SARP	23	34	43	15	36	44
BB-SARP	0	0	0	8	10	10
PNB	49	59	71	31	58	69
PNB-PE	51	63	75	39	67	76
Number of subjects		99			102	

Note: Entries count the number of subjects who pass each test at the listed error allowance.

point predictions in D2 and D4. Random behavior has a 0.001 percent chance or less of passing either BB- or NB-SARP with 1 error and a 0.3 percent chance of passing the PNB Algorithm with 1 error (see online Appendix D). Table 4 shows how many subjects pass each test.

We compare results of the tests allowing for up to one error. We find that no subjects pass BB-SARP in Risk, and only 10 percent of subjects pass it in Social. However, 34–35 percent of subjects in each experiment pass NB-SARP. While fewer subjects pass BB-SARP than either BB-WARP or BB-Mon, a similar number of subjects pass all NB-WARP restrictions with one error and the more demanding NB-SARP with two errors. These results show that a plurality of our subjects are well described as narrow bracketers.

Our test of partial-narrow bracketing diagnoses how many of those who fail BB-SARP and NB-SARP behave consistently with intermediate degrees of bracketing. We find that 15 percent of subjects in Risk and 12 percent of subjects in Social pass the PNB test but neither BB-SARP nor NB-SARP when allowing for 1 error.[24] Only 4 percent of subjects in Risk and 9 percent of subjects in Social pass the PNB-PE test but not the PNB test.

Result 2: When allowing for 1 error relative to each test, 34 percent and 35 percent of subjects are rationalizable by narrow bracketing, and 0 percent and 10 percent of subjects are rationalizable by broad bracketing in the Risk and Social Experiments, respectively. An additional 25 percent and 12 percent of subjects are rationalizable by partial-narrow bracketing but not narrow or broad bracketing.

We note briefly that a subject whose underlying preferences are symmetric and linear would make the same choices regardless of how they bracket.[25] They would be indifferent between many bundles in our experiment and so violate our tests' assumption that choices reveal strict preferences. However, this cannot bias our

[24] Broad and narrow bracketing are both special cases of partial-narrow bracketing, and thus, any subject who passes BB- or NB-SARP will also pass this test.

[25] Symmetric linear preferences require only that $x_A^{1,1} = x_{A/C}^{3,2} = x_A^{5,1} = 0$. Linear preferences are the only class of preferences consistent with both narrow and broad bracketing (Theorem 4 in Appendix A).

Table 5—Shopping Tests

	D1			D3			Both			Full		
No. of errors	0	1	2	0	1	2	0	1	2	0	1	2
NB	23	23	60	53	65	69	20	21	49	15	16	40
BB	21	22	27	23	24	25	13	15	18	12	14	16
PNB	45	46	95	76	91	98	33	37	68	27	30	56
PNB-PE	45	46	95	76	93	98	33	37	70	27	30	57
No. of subjects	101											

Note: Entries count the number of subjects who pass each test at the listed error allowance.

conclusions too much since only five subjects in Risk and two subjects in Social could possibly be consistent with it.[26]

C. *Shopping Experiment: Induced-Value Tests*

The tests of our predictions thus far assume that utility is not observed. When utility is known, as in our Shopping Experiment, narrow and broad bracketing each make unique predictions in each decision.[27] To test the models, we compare how far each subject's choices are from each model's predictions (Table 5).

Testing the point predictions of narrow bracketing in each of Decisions 1 and 3, 23 percent and 64 percent of subjects are respectively within 1 error of the predictions of narrow bracketing, while 21 percent are consistent in both. Allowing for 2 errors raises pass rates to 59 percent for Decision 1 and 49 percent for both.[28] Looking at the full set of implications of narrow bracketing on all choices made in the experiment, 40 percent of subjects are within 2 errors of passing.

In contrast, 21 percent, 24 percent, and 14 percent of subjects are within 1 error of being consistent with broadly bracketed maximization in Decisions 1, 3, and both, respectively. When allowing for two errors, those numbers remain similar. Using all decisions in the experiment, only 16 percent of subjects are within two errors of being consistent with all implications of broad-bracketed maximization. PNB describes less than 2 percent of the remaining subjects.

Since we induce the payoff function, we are able to compute the value of α that best fits a subject's behavior (up to the limits imposed by discretization of the budget sets) under the assumption that the induced-value function acts as their utility function. To

[26] One subject in each of Risk and Social passes NB-SARP, and the other subject in Social passes BB-SARP when allowing one error. Two other subjects in Risk also had $x_C^{1,2} = x_A^{3,1} = 0$ and so pass BB-WARP, BB-Mon, and NB-WARP but neither NB-SARP nor BB-SARP. In total, one in each of Risk and Social is classified as broad, two in Risk and one in Social are classified as narrow, and the remaining two in Risk are unclassified.

[27] Narrow-bracketed maximization implies $x_a^{1,1} = 1$, $x_a^{1,2} = 6$, $x_a^{3,1} = 5$, and $x_a^{3,2} = 4$. Broad-bracketed maximization implies $x_a^{1,1} = 0$, $x_a^{1,2} = 10$, $x_a^{3,1} = 10$, and $x_a^{3,2} = 0$.

[28] The sharp difference between one- and two-error tests in Decision 1 but not Decision 3 results from the fact that 40 subjects selected $x_a^{1,1} = 2$, $x_o^{1,1} = 4$, which is the second-best available bundle from a narrow bracketer's perspective but is two lines away from the best bundle of $x_a^{1,1} = 1$, $x_o^{1,1} = 6$—due to the discreteness of the budget set, one apple and five oranges is between these bundles. We note that this deviation is in the opposite direction of that predicted by broad or partial-narrow bracketing, and 32 of these subjects also selected $x_a^{1,2} = x_o^{1,2} = 6$.

that end, we compute the point predictions of the partial-narrow bracketing model for each $\alpha \in \{0, 0.01, 0.02, \ldots, 0.99, 1\}$ for Decisions 1 and 3 and obtain distinct predictions for nine intervals of α. We assign each subject to the range of α for which their choices exhibit the fewest errors relative to that range's predictions. We find that 64 percent of subjects are classified to a range that includes full narrow bracketing, $\alpha = 1$, and 25 percent are classified to a range that includes full broad bracketing, $\alpha = 0$. Of the remaining subjects, none are best described by $\alpha \in [0.25, 0.71]$. This suggests that even those subjects who are not exactly described by either broad or narrow bracketing are close.[29]

D. *Classifying Subjects to Models*

The tests thus far do not make any adjustment for the fact that partial-narrow bracketing nests narrow and broad bracketing as polar cases and can thus accommodate more behavior. To compare the predictive success of each model at the subject level, we use a subject-level implementation of the Selten score (Selten 1991; Beatty and Crawford 2011). For each subject and each model (symmetric versions for Risk and Social, using the induced-value function for Shopping), we calculate the number of errors the subject exhibits relative to that model. Then, we calculate the number of possible choice combinations in the experiment that are consistent with that model and that number of errors—the model-error pair's "predictive area." We divide the predictive area by the total number of possible combinations of choices in the experiment to compute the measure for each subject i and model $m \in$ {broad, narrow, PNB, PNB-PE} as $\text{predictive\_success}_{i,m} = 1 - \frac{\#\text{predictive\_area\_for\_}i,m}{\#\text{all\_possible\_choices}}$.[30] We use all choices made in the experiment to assign each subject to the model with the highest predictive success; in cases where every rationalizing model-error pair for a subject would rationalize more than 1 million possible combinations of choices in our experiment, we categorize them as "Unclassified."

Across the three experiments, we classify 67–78 percent of subjects as narrow bracketers (Table 6). In contrast, 2 percent, 10 percent, and 27 percent of subjects are classified as broad bracketers in the Risk, Social, and Shopping Experiments, respectively. However, 7 percent, 6 percent, and 5 percent were classified to one of the two partial-narrow bracketing models. After adjusting for predictive power, partial-narrow bracketing does not help explain very many subjects' behavior.

To form confidence intervals, we assume that model of bracketing is multinomially distributed with fixed strike rate within each treatment. We calculate confidence intervals according to the Wilson (1927) score. In all three treatments, the lower

[29] Because of discreteness, $[0,1]$ can be partitioned so that the parameters in each cell make the same choices. In this case, the partition is $\{[0, 0.04], [0.05, 0.06], [0.07, 0.14], [0.15, 0.24], [0.25, 0.29], [0.3, 0.38], [0.39, 0.71], [0.72, 0.75], [0.76, 1]\}$.

[30] In the Risk and Social Experiments, there are $(11 \times 17) \times 15 \times (11 \times 11) \times 17 \times 11 = 63{,}468{,}735$ possible combinations of choices. Symmetric narrow bracketing allows 6, 87, and 606 possible combinations of choices when allowing for 0, 1, and 2 errors, respectively, whereas symmetric broad bracketing allows 12, 116, and 585 combinations of choices; symmetric PNB allows 35,797, 200,828, and 597,728 combinations; and symmetric PNB-PE allows 116,267, 619,375, and 1,725,466 combinations. Thus, a subject whose choices are consistent with partial-narrow bracketing will be classified as a partial-narrow bracketer if and only if they are sufficiently far from being consistent with both broad and narrow bracketing.

Table 6—Classification of Subjects

	Percent Selten score maximized		
	Risk	Social	Shopping
Broad bracketing	2.02 (0.55, 7)	9.8 (5.52, 17.46)	26.73 (19.26, 36.46)
Narrow bracketing	77.78 (67.95, 83.99)	75.49 (67.64, 84.47)	67.33 (58.27, 76.45)
PNB	7.07 (3.43, 13.74)	1.96 (0.55, 7.01)	3.96 (1.57, 9.84)
PNB-PE	0 (0, 3.7)	3.92 (1.57, 9.85)	0.99 (0.18, 5.45)
Unclassified	13.13 (7.76, 20.97)	8.82 (4.81, 16.24)	0.99 (0.18, 5.45)

Note: To calculate confidence intervals, we assume that the model of bracketing is multinomially distributed with fixed strike rate within each treatment and calculate the Wilson (1927) score.

bound of the interval for narrow bracketers exceeds the upper bound of the intervals for all other models of bracketing. Most other intervals overlap, but there are significantly more broad bracketers than partial-narrow bracketers in the Shopping Experiment.

Result 3: Judging each model's fit by its predictive success, 67–78 percent of subjects are classified as narrow bracketers, 2–27 percent are classified as broad bracketers, and 5–7 percent are classified as partial-narrow bracketers across the three experiments.

V. Robustness and Secondary Analyses

To study how choice architecture mediates bracketing, we conducted an online version of our Risk Experiment that varied the presentation in a two-by-two design. First, the "Examine" treatment instructed the subject to "First, examine both accounts, then purchase your investments" in the instructions, tested this in a quiz question, then included that text at the top of each two-part decision screen; the "Basic" treatment did not. Second, the "Tabs" treatment presented each part of a decision as a separate HTML tab (analogous to our separate pages in our paper experiments), whereas the "Side-by-Side" treatment presented parts of a decision side by side on the same screen (analogous to Tversky and Kahneman 1981). We recruited 200 US-based subjects from Prolific Academic to participate and randomly assigned each to one of the four treatment pairs.

We also replicated our Shopping Experiment online with 46 subjects from Prolific Academic. For all 46, we implemented the Side-by-Side and Examine interventions, eliminated the quantity-restricted sale in D2, and provided subjects with a calculator instead of a payoff table. We provide detailed screenshots and results in the online Appendix.

In Section VA, we use the online experiments to argue that presentation has a limited effect on our results. The online experiments allow us to collect nonchoice data that shed light on the subjects' choice processes. In Section VD, we use these data to argue that a substantial fraction of narrow bracketers give some consideration to both parts of the decision before choosing. We then explore the robustness of our tests of broad bracketing to extremeness aversion, investigate the possibility of order effects or learning, and perform statistical tests of the differences we find across the domains.

A. *Effects of Choice Architecture*

The online subject pool is more slanted toward narrow bracketing than our original pen-and-paper study. Pass rates for each NB-WARP test are higher, and pass rates for BB-WARP and BB-Mon are generally lower. Only 2 of 200 (1 percent) subjects pass BB-SARP when allowing for 2 errors, and these 2, and only these 2, are classified as broad bracketers according to Selten score. In contrast, 96 (48 percent) subjects pass NB-SARP when allowing 1 error, and 84 percent are classified as narrow bracketers by Selten score.

We find almost no effect on choices from either of the two interventions in the online Risk Experiment. Both subjects who pass BB-SARP are in the Examine and Tabs treatment, contrary to our expectation that Side-by-Side would be more conducive to broad bracketing. Neither treatment has a large or statistically significant (with a small sample size caveat) effect on the rate of broad bracketing ($p = 0.22$, $p = 0.24$, for Fisher's exact tests of the Examine and Tabs treatments, respectively). Within the Tabs group, Examine does not have a statistically significant effect ($p = 0.20$, Fisher's exact test). We similarly find no effect of the treatments on the rates of narrow bracketing ($p = 1.00$ for both, Fisher's exact tests).

In addition, the online Shopping Experiment implemented both Examine and Side-by-Side. It provided subjects with a calculator instead of a payoff table. We classified 5 out of 46 (10.86 percent) subjects to broad bracketing and 33 (71.74 percent) to narrow. The rate of narrow bracketing did not change much, while the rate of broad bracketing slightly decreased relative to the pen-and-paper Shopping Experiment ($p = 0.03$, Fisher's exact test).

All in all, this suggests that the low rates of broad bracketing we find are not overly sensitive to varying the choice architecture to encourage broad bracketing. We caution against overinterpreting this result since broad bracketing was so rare in all treatments. The shift to an online interface and the Prolific subject pool may have had more effect than any of the nudges. We suspect that more extreme nudges and decision aids might be more effective.

B. *Nonchoice Data and Bracketing*

Our online experiments shed light on how the choice process, and in particular consideration, differed across subjects. We have two tools for measuring consideration. In the Tabs arm of online Risk, we observe which Tabs subjects clicked and when they made their decision. In online Shopping, we record how subjects used the calculator. These nonchoice data show that only some narrow bracketers completely ignore other parts of the decision. In both experiments, more than a quarter of narrow bracketers gave some consideration to both parts of the decision. These subjects had enough information to bracket more broadly yet did not.

In the Tabs versions of the online Risk Experiment, we record whether each subject clicked on both tabs before making their final choices. Only 28 of 102 subjects did so in both D1 and D3. While this includes the 2 broad bracketers, the other 26 subjects had sufficient information to bracket broadly but failed to do so. Of the 28, 22 were in the Examine-Tabs treatment, and 6 were in the Basic-Tabs treatment ($p < 0.01$, Fisher's exact). The prompt was effective at increasing this click pattern, but it did not significantly affect the classification to broad bracketing ($p = 0.20$, Fisher's exact). Clearly, paying attention to both parts of a decision is necessary to bracket broadly. However, the fraction of narrow bracketers among those who paid attention was not substantially different than the fraction among those who did not (21 of 28 versus 61 of 74, $p = 0.41$, Fisher's exact). At the other extreme, we find that 33 subjects made their final choice in each of D1 and D3 before having seen both parts and thus could not have been broad bracketing; 29 of these subjects are classified as narrow bracketers.

While intensity of calculator use does not differ much between broad and narrow bracketers in the online Shopping Experiment, patterns of calculator use do.[31] Ex ante, we expected that plugging in bundles that are feasible in the decision as a whole (e.g., (9, 11) for D3) to the calculator (a "broad calculation") would predict broad bracketing and that plugging in a bundle that is available in a part of a decision but lies below the decision-level feasible set (e.g., (5, 5) for D3) to the calculator (a "narrow calculation") would predict narrow bracketing. We classify nine of ten subjects who made a narrow calculation as narrow bracketers. Surprisingly, we classify 9 of 16 subjects who made at least one such "broad calculation" as narrow bracketers and only 3 as broad. These subjects appear to actively attend to how the parts fit together yet still make narrowly bracketed choices.

In summary, the nonchoice data suggest diversity in the relationship between consideration and bracketing. It suggests that about a quarter of narrow bracketers appear to follow a choice process that involves only considering one part at a time, but these same data also show that a quarter or more of narrow bracketers gave some consideration to both parts. The seemingly weak link between broader consideration and broadly bracketed choices suggests why the interventions we designed to encourage broad bracketing in the online Risk Experiment were not very effective. However, the sparsity of broad bracketers in both experiments makes it hard for us to tell what drives it. The choice processes leading to narrow or broad bracketing are a fertile direction for future work.

C. *Effects of Extremeness Aversion*

As we describe in Section I, broad bracketing requires a corner solution in at least one part of any multipart decision. Evidence from other contexts suggests that some subjects may be *extremeness averse* and avoid corner choices. For example, subjects in linear public good games tend not to play the Nash strategy of making no contributions. However, in nonlinear public good games where the Nash strategy requires a positive but not complete contribution, subjects play the equilibrium strategy more

[31] The average (median) number of calculations was 10.40 (12) for broad bracketers, 12.21 (8) for narrow bracketers, and 12.86 (10) for all subjects.

frequently (see, e.g., Section 2 of Vesterlund 2016 for a discussion). Extremeness aversion could affect our conclusions about broad bracketing and its prevalence relative to narrow bracketing.

Nonextreme allocations in two-part decisions lead to violations of BB-Mon. However, few additional subjects are consistent with a relaxation of BB-Mon that allows for subjects to be close to, but not necessarily at, a corner when required. Formally, we say that a subject passes *Extremeness-Averse* (*EA*-)*BB-Mon* if they are within two tokens of making an extreme allocation in each decision required by BB-Mon. Without allowing for any errors, three subjects in Risk and zero subjects in Social pass EA-BB-Mon but not BB-Mon. Allowing for 2 errors, 14 percent of subjects in Risk and 3 percent in Social pass EA-BB-Mon but not BB-Mon. As a benchmark, 26 percent of all possible datasets pass EA-BB-Mon but not BB-Mon when allowing for 2 errors.

Extremeness aversion is unlikely to affect our classification or tests. Of the 77 subjects classified as narrow bracketers in each of Risk and Social, none pass EA-BB-Mon but not BB-Mon (allowing for 2 errors, this increases to 6 and 1, respectively). No subject in either Risk or Social passes both NB-SARP and EA-BB-Mon but not BB-Mon when allowing for no or one errors in each test. Allowing two errors in each test, only three subjects in Risk and one in Social pass both NB-SARP and EA-BB-Mon but not BB-Mon. Since BB-Mon is a necessary but not sufficient condition for broad bracketing, a highly risk-tolerant or inequity-neutral narrow bracketer would pass both NB-SARP and BB-Mon. Only one subject in each does so. Somewhat more risk- or inequity-averse narrow bracketers will also pass EA-BB-Mon.

This suggests that the effect of extremeness aversion on our tests is probably not too large. In addition, a similar number of subjects are consistent with broad bracketing in decisions that require two corner choices as are consistent in decisions that require only one corner choice.[32] The evidence still suggests heterogeneity in bracketing, with a plurality best described as narrow bracketers, even after adjusting for extremeness aversion.[33]

D. *Welfare Losses and Measurement*

So far, we have focused on identifying how our subjects bracketed. We now apply our classification to quantify in dollar terms the payoffs forgone by narrow bracketers (Table 7). We also illustrate how looking at aggregate data can lead to misleading inferences. A narrow bracketer's preference parameters should be measured using parts as the unit of observation, and a broad bracketer's should use decisions as the unit of observation. Using the wrong unit of observation, necessary for an aggregate

[32] A broad bracketer in Shopping buys only apples in D3.1, only oranges in D3.2, and only oranges in D1.1 but buys both fruits in D1.2. D3 requires two extreme choices, and D1 requires only one. As reported in Table 5, there are a very similar number of subjects close to these predictions in D1 and D3.

[33] We also conducted a follow-up experiment for which only one of the two treatments requires a corner choice. To get more separation between narrow and broad, we used a much finer choice grid for each budget. Unfortunately, this made the data too noisy to draw strong conclusions. No more than 14.8 percent of subjects are close to the predictions of either narrow or broad bracketing combined in either treatment. Nonetheless, the ratio of broad to narrow bracketers does not vary much across treatments, lending weak support to our conclusion. We elaborate on this experiment in online Appendix J.

Table 7—Losses and Payoff Differences by Parts and Decisions

	D1.1	D1.2	D1	D3.1	D3.2	D3	D5	*n*
Panel A. Payoff loss ($)								
Risk	0.39	0	0.39	0	0.35	0.35	0.38	99
NB	0.4	0	0.4	0	0.38	0.38	0.41	77
BB	0	0	0	0	0.2	0.2	0.3	2
Social	0.47	0	0.47	0	0.46	0.46	0.49	102
NB	0.51	0	0.51	0	0.51	0.51	0.49	77
BB	0	0	0	0	0.03	0.03	0.44	10
Shopping	0.56	0.51	1.06	1.03	0.85	1.25	0.05	101
NB	0.22	0.07	1.41	0.01	0.07	1.67	0.03	68
BB	1.23	1.31	0.35	3.76	2.94	0.14	0.04	27
Panel B. Payoff difference ($)								
Risk	4.24	2.24	5.39	1.88	4.7	4.51	3.79	99
NB	3.47	0.91	3.91	0.7	3.72	3.98	3.24	77
BB	12	9	3	3	7.6	4.6	6.6	2
Social	3.1	2.39	1.54	2.29	3.35	1.91	2.58	102
NB	1.64	0.52	1.62	0.6	1.68	1.68	2.17	77
BB	12	12	0	10	11.34	1.46	2.56	10

Notes: Panel A: "Payoff loss" columns report total $ loss (Shopping), loss in expected value (Risk), and average loss per recipient (Social) compared to expected/average value maximization. We evaluate each part on its own in columns D1.1 and D3.2 and integrate across parts in columns D1 and D3. In D1.2 and D3.1 all allocations have no loss in either Risk or Social. Panel B: "Payoff difference" columns report the difference in $ allocations across the two states or people.

analysis with heterogeneous bracketing, leads to misleading conclusions about preferences. We focus on the Social and Shopping Experiments because there are only two participants classified as broad bracketers in Risk.

In the Shopping Experiment, we can precisely measure welfare losses through the earnings that participants did not receive.[34] Narrow bracketers had lower earnings than broad bracketers in both two-part decisions. They had $1.41 and $1.67 in forgone payments compared to just $0.35 and $0.14. The differences between the maximum and the minimum payoffs are $9.60 and $10.36 in D1 and D3, so the payoff losses for narrow bracketers are 10.9 percent and 14.8 percent of the variable payment. In the one-part decision D5, narrow and broad bracketers should make identical choices. They do indeed make very similar choices, suggesting that arithmetic errors do not explain the difference. Indeed, narrow bracketers do a good job at optimizing part by part and would forgo no more than $0.22 if the compensation was based on individual parts of a decision rather than the decision as a whole.[35]

The "Payoff difference" columns of Table 7 illustrate how it is easy to obtain misleading inferences about how subjects make the equity-efficiency trade-off. In the one-part decision D5, narrow bracketers implemented allocations that differed by an average of $2.17 between A and B, while the corresponding average difference

[34] Interpreting payoff losses as welfare losses requires that broad bracketing does not have a direct utility cost.

[35] There is a similar pattern for expected or aggregate payoffs in the Risk and Social Experiments, but risk or inequity aversion makes interpreting this as a welfare loss more ambiguous.

is \$2.56 for broad bracketers ($p = 0.18$, rank sum test comparing D5 allocations). Narrow and broad bracketers make similar trade-offs between equity and efficiency from one-part decisions. Yet if we look at D1.2 or D3.1 on its own, ignoring the other part of the decision, we would incorrectly infer that broad bracketers care mainly about efficiency, while narrow bracketers make near-equal allocations on average, consistent with inequity aversion. We would obtain an opposite conclusion by looking at D1 as a whole, where narrow bracketers' decision-level allocations involve some inequity on average, but all broad bracketers make perfectly equal final allocations.

E. *Differences across Experiments*

The fraction of subjects classified to broad bracketing varies across experiments, from 1 percent for online Risk to 27 percent in pen-and-paper Shopping. There is a significant difference in broad bracketing rates between pen-and-paper Shopping and each of Risk and Social ($p < 0.01$, Fisher's exact test) as well as between the online and pen-and-paper Shopping Experiments ($p = 0.03$).[36] However, narrow bracketing rates varied much less, from 67 percent to 78 percent across pen-and-paper experiments, with no significant differences in these rates ($p > 0.10$ for all pairwise Fisher's exact tests). Nor was there a significant difference in narrow bracketing rates between the online and pen-and-paper versions of Risk and Shopping ($p > 0.10$ for both Fisher's exact tests). Explanations for the higher rate of broad bracketing in Shopping and lower rate in Risk include the more naturalistic setting, the presence of an objectively correct payoff function, and cognitive difficulties specific to choice under risk (as suggested by Martínez-Marquina, Niederle, and Vespa 2019). We cannot distinguish between these explanations.

VI. Conclusion

We propose revealed preference tests for how a person brackets their choices that rely only on monotonicity of underlying preferences. We deploy these tests in an experiment where both narrow and bracketing make falsifiable predictions, unlike in past work. Across our experiments, at least twice as many subjects were classified as narrow bracketers than as broad bracketers. A majority of people tend to narrowly bracket, while a noticeable minority broadly bracket. While many of our subjects are not well described by either broad or narrow bracketing, our novel tests of partial-narrow bracketing suggest that it does not do much better after adjusting for predictive power. This suggests that applications should calibrate a population mix of broad and narrow bracketers rather than a representative agent model with a calibrated partial-narrow bracketing parameter (as in Barberis and Huang 2007).

Bracketing rates differ across tasks, domains, and subject pools. While our framework is well suited to detect and to measure these differences, it is less well suited

[36] The difference between online and pen-and-paper Risk was not significant, but this is to be expected given the negative effect on broad bracketing and the already small numbers of broad bracketers.

to determine why these differences persist. Nonchoice data, as we collected in the online follow-up experiment, can help. It seems to rule out uniform explanations for why so many bracket narrowly, such as lack of awareness of complementarities across parts. We think understanding why people bracket the way they do is an interesting direction for future work.

Appendix A. Theoretical Appendix: Proofs, Derivations, and PNB Algorithm

PROOF OF THEOREM 1:

One can follow the usual proof that SARP holds if and only if a dataset is rationalizable by a complete and transitive preference relation to establish the result with a preference relation; see, e.g., Mas-Colell, Whinston, and Green (1995). Establishing rationalization with a utility function requires a bit more work. We provide an argument below, as we are unaware of one in the literature.

Given $\mathcal{O} = (x^i, B^i)_{i=1}^N$ for either $\mathcal{O} = \mathcal{D}^{NB}$ or $\mathcal{O} = \mathcal{D}^{BB}$, set $X' = \cup_i B^i$. Observe that each B^i is a compact set, either by assumption (for $\mathcal{O} = \mathcal{D}^{NB}$) or since the finite sum of compact sets is compact (for $\mathcal{O} = \mathcal{D}^{BB}$). Thus, X' is also compact. It is therefore bounded as a subset of $\mathbb{R}^n$. Let X be a closed ball centered at zero containing X'.

Let $\succsim$ be the transitive closure of $P^{\mathcal{O}}$ defined on X. We show that $\succsim$ is closed as a subset of $X \times X$. First, note $P^{\mathcal{O}}$ itself is closed as a finite union of the closed sets $\{(y,z) \in X \times X : y \geq z\}$ and $\{x^i\} \times B^i$ for all $i = 1, \ldots, N$. Second, we show that $y \succsim z$ if and only if there exist $y_1, y_2, \ldots, y_{K-1}, y_K$ so that $y_i P^{\mathcal{O}} y_{i+1}$ for all i, $y_1 = y$, and $y_K = z$, where $K \leq 2N + 1$. Consider any $y \succsim z$. By definition, there exist $y_1, y_2, \ldots, y_K$ so that $y_i P^{\mathcal{O}} y_{i+1}$ for all i, $y_1 = y$, and $y_K = z$ and there is no shorter sequence that establishes $y \succsim z$. If $y_j \geq y_{j+1}$ and $y_{j+1} \geq y_{j+2}$ for some j then $y_j \geq y_{j+2}$, and we can construct a shorter sequence by leaving y_{j+1} out. If $y_j \not\geq y_{j+1}$, then $y_j = x^{i(j)}$ and $y_{j+1} \in B^{i(j)}$ for some $i(j)$. Then, $y_k \neq y_j$ for all $k > j$ by SARP, so at most one index j' has $i(j') = i$ for each i. Therefore, the longest possible shortest sequence alternates $\geq$ with direct revelations from choice. There can be no more than $2N + 1$ of these, so $K \leq 2N + 1$. Moreover, since we can trivially append $y_{K+1} = y_K$ to the end of such sequences, we can consider only sequences with exactly $2N + 1$ in what follows.

Now, pick any sequence $(y_m, z_m)_{m=1}^{\infty}$ contained in $\succsim$ that converges to (y,z). Then, there exists $y_m^2, \ldots, y_m^{2N}$ so that $y_m P^{\mathcal{O}} y_m^2 P^{\mathcal{O}} \ldots P^{\mathcal{O}} y_m^{2N} P^{\mathcal{O}} z_m$. The sequence $(y_m, y_m^2, \ldots, y_m^{2N}, z_m)$ has a convergent subsequence $(y_{m_k}, y_{m_k}^2, \ldots, y_{m_k}^{2N}, z_{m_k})$ since $X \times \ldots \times X$ is compact by the Tychonoff Theorem. Let $(y^*, y^{2*}, \ldots, y^{2N*}, z^*)$ be its limit. Since $P^{\mathcal{O}}$ is closed, $y^* P^{\mathcal{O}} y^{2*} P^{\mathcal{O}} \ldots P^{\mathcal{O}} y^{2N*} P^{\mathcal{O}} z^*$. Finally, $(y^*, z^*) = (y,z)$ since $(y_m, z_m) \to (y,z)$. Hence, $\succsim$ is closed.

Now, apply Levin's (1983) Theorem, as in Nishimura, Ok, and Quah (2017), to get a utility function U from X to $[0,1]$ so that $x \succ y$ implies $U(x) > U(y)$ and $x \succsim y$ implies $U(x) \geq U(y)$. Since $\geq$ is included in $\succsim$, $U(\cdot)$ is increasing. Since $x^i \succ y$ for all $y \in B^i \backslash \{x^i\}$, $U(x^i) > U(y)$ for all $y \in B^i \backslash \{x^i\}$. Extending to all of $\mathbb{R}_+^n$ is straightforward: simply set $U(y) = V(y)$ whenever $y \notin X$ for an arbitrary increasing function $V(\cdot)$ picked so that $V(y) > 1$ for all y. ∎

PROOF OF THEOREM 2:

Fix α and $\mathcal{D}$. For every decision t, define the lottery

$$p^t = \frac{\alpha K_t}{\alpha K_t + (1-\alpha)}\left(\frac{1}{K_t}, x^{t,k}\right)_{k=1}^{K_t} + \frac{(1-\alpha)}{\alpha K_t + (1-\alpha)}(1, x^t),$$

and $Y \subseteq \mathbb{R}^n$ to be a finite set that includes the union of the supports of $p^1, \ldots, p^T$ and the vector zero. It will be convenient to take Y equal to this set, but this is inessential. Denote by ΔY the finite support lotteries over Y. Consider the set

$$Q_Y^t = \left\{\frac{\alpha K_t}{\alpha K_t + (1-\alpha)}\left(\frac{1}{K_t}, y^k\right)_{k=1}^{K_t} + \frac{(1-\alpha)}{\alpha K_t + (1-\alpha)}(1, y) \in \Delta Y : \exists z^k \in B^{t,k} \text{ so that } y^k \le z^k \text{ and } y \le \sum z^k\right\}$$

is a finite set of lotteries over Y that includes p^t and any others that are affordable. The ancillary dataset

$$\mathcal{D}_Y^\alpha = \left\{\left(p^t, Q_Y^t\right)\right\}_{t=1}^T$$

is strictly rationalizable by expected utility if and only if $\mathcal{D}$ is rationalizable by α-partial-narrow bracketing.

If so, then there exists an EU preference V over ΔY so that

$$V(p^t) > V(q^t) \; \forall\, q^t \in Q_Y^t \setminus p^t$$

for all t. Let v be the utility index of V and extend to $\mathbb{R}_+^n$ by

$$u^*(z) = \max_{y \le z,\, y \in Y} v(y).$$

Then for any t and $y^{t,1} \times \ldots \times y^{t,K_t} \in B^{t,1} \times \ldots \times B^{t,K_t} \setminus \left\{x^{t,1} \times \ldots \times x^{t,K_t}\right\}$,

$$\alpha \sum_k u^*(x^{t,k}) + (1-\alpha)\, u^*(x^t) = \kappa_t V(p^t) > \kappa_t \max_{q^t \in Q_Y^t \setminus p^t} V(q^t)$$

$$\ge \alpha \sum_k u^*(y^{t,k}) + (1-\alpha)\, u^*\left(\sum_{k=1}^{K_t} y^{t,k}\right),$$

where $\kappa_t = \alpha K_t + (1-\alpha)$.[37] To find a strictly increasing u, let u^{**} be a strictly increasing extension of v. Choosing $\epsilon > 0$ small enough, for $u = (1-\epsilon)\, u^* + \epsilon u^{**}$,

$$\alpha \sum_k u(x^{t,k}) + (1-\alpha) u(x^t) > \alpha \sum_k u(y^{t,k}) + (1-\alpha) u\left(\sum_{k=1}^{K_t} y^{t,k}\right)$$

[37] The max exists since Q_Y^t is a finite set.

for all t and any $y^{t,1} \times \ldots \times y^{t,K_t} \in B^{t,1} \times \ldots \times B^{t,K_t} \setminus \{x^{t,1} \times \ldots \times x^{t,K_t}\}$, establishing that $\mathcal{D}$ is rationalized by α-PNB.[38]

Conditions under which $\mathcal{D}_Y^\alpha$ is rationalized by expected utility are well-known. Here, we follow Clark (1993). As standard, p is directly revealed preferred to q, written $p \succ q$, if $p = p^t$ and $q \in Q_Y^t \setminus \{p^t\}$ for some decision t. Define $p \succsim q$ if $p \succ q$ or $p = q$. We say that p is indirectly reveal preferred via independence to q, written $p \,\widetilde{\succsim}\, q$, whenever there exist $r, p_1, \ldots, p_n, q_1, \ldots, q_n \in \Delta Y$, $\beta \in (0,1]$, and $\lambda_1, \ldots, \lambda_n > 0$ so that

$$\beta p + (1-\beta) r = \sum_{i=1}^{n} \lambda_i p_i$$

$$\beta q + (1-\beta) r = \sum_{i=1}^{n} \lambda_i q_i$$

$$p_i \succsim q_i \quad \text{for all} \quad i = 1, \ldots, n$$

$$\sum_{i=1}^{n} \lambda_i = 1$$

and $p \,\widetilde{\succ}\, q$ holds whenever at $p_i \succ q_i$ for some i. The linear axiom of revealed preference (LARP) is that $q \,\widetilde{\succsim}\, p$ implies that $p \,\widetilde{\succ}\, q$ does not hold. Theorem 3 of Clark (1993), combined with the finiteness of $\mathcal{D}^\alpha$, implies that $\widetilde{\succsim}$ has an expected utility rationalization. Conclude $\mathcal{D}^\alpha$ satisfies LARP if and only if $\mathcal{D}$ is rationalized by α-partial-narrow bracketing.

The following algorithm inputs $\mathcal{D}^\alpha$ and outputs one if and only if $\mathcal{D}$ is rationalized by α-partial-narrow bracketing for any rational number $\alpha \in [0,1]$. For concreteness, label $Y = \{y_1, y_2, \ldots, y_m\}$. We identify each $p \in \Delta Y$ with the $p \in \mathbb{R}^m$ so that $p_i = p(y_i)$.

Define the matrix A with a column equal to $p^t - q$ for each $q \in Q_Y^t \setminus p^t$ and every t. A is a $D \times m$ matrix interpreted as strict preference, where D is the number of data points extracted. Let $\vec{1}$ be a $D \times 1$ matrix of ones and

$$\hat{A} = \begin{bmatrix} A \\ \vec{1} \end{bmatrix}.$$

Let $\hat{b}$ be an $m+1$ vector with the first m components zero and the last one.

Note that if $p \,\widetilde{\succsim}\, q$, then there exists $\beta > 0$, $\lambda_i \geq 0$, and $p_i \succsim q_i$ for $i = 1, \ldots, m$ so that

$$\beta(p - q) = \sum_{i=1}^{m} \lambda_i (p_i - q_i)$$

and $\sum_{i=1}^{m} \lambda_i = 1$. Each $p_i - q_i$ corresponds to a column of A, so this is equivalent to $p - q \in C = cone\left(co\left(\{A_1, \ldots, A_D\}\right)\right)$, where A_i is the ith column of A. LARP holds when $p - q \in C$ implies $q - p \notin C$, or, equivalently, if and only if $0 \notin C$.

[38] This approach is similar to that of Polisson, Quah, and Renou (2020).

That is, there does not exist $\phi^1, \ldots, \phi^D \geq 0$ so that $\sum \phi^i = 1$ so that $\sum_{i=1}^{D} \phi^i A_i = 0$, which is equivalent to $\hat{A}\phi = \hat{b}$.[39]

Solve the system

$$\min_{y \in \mathbb{R}^{m+1}} y^T \hat{b}$$

$$\text{subject to} \quad \hat{A}^T y \geq 0$$

for y. The Ellipsoid algorithm provides a polynomial-time solution to this. Let y^* be the solution, if one exists. There are two cases.

Case 1: There is no solution to $\hat{A}^T y \geq 0$, or y^* satisfies $y^{*T}\hat{b} \geq 0$. Then, Farkas's Lemma implies there exists $\phi \geq 0$ so that $\hat{A}\phi = \hat{b}$. LARP fails by the above, so return zero.

Case 2: The program is unbounded below, or $y^{*T}\hat{b} < 0$. Then, Farkas's Lemma implies there *does not* exist any $\phi \geq 0$ so that $\hat{A}\phi = \hat{b}$. LARP is satisfied by the above, so return one.

To specialize the above to PNB-PE, we perform the analysis part by part rather than decision by decision. Let $p^{t,k} = \left(\alpha, x^t; (1-\alpha), x^{t,k}\right)$, $Y \subseteq \mathbb{R}^n$ be a finite set that includes the union of the supports of the lotteries $p^{t,k}$, and

$$Q_Y^{t,k} = \left\{\left(\alpha, y; (1-\alpha), y'\right) \in \Delta Y : \exists x \in B^{t,k} \text{ so that } y' \leq x + x^{t,-k} \text{ and } y \leq x\right\}.$$

Repeating the above with the ancillary dataset

$$\mathcal{D}_Y^{PE,\alpha} = \left\{\left(p^{t,k}, Q^{t,k}\right)\right\}_{(t,k)}$$

replacing $\mathcal{D}^\alpha$ establishes the result. ∎

THEOREM 3 (Identification of α): S*uppose that u is a known differentiable and strictly quasi-concave function and $\mathcal{D}$ is rationalized by partial-narrow bracketing. Further suppose that for some t,k, $B^{t,k}$ is a Walrasian budget set for prices $p^{t,k}$, and $n = 2$, $\frac{p_1^{t,1}}{p_2^{t,1}} \neq \frac{p_1^{t,2}}{p_2^{t,2}}$, $\lim_{x_1 \to 0^+} \frac{\partial u(x_1,x_2)}{\partial x_1} = +\infty$ for all $x_2 > 0$, and $\lim_{x_2 \to 0^+} \frac{\partial u(x_1,x_2)}{\partial x_2} = +\infty$ for all $x_1 > 0$.*

Then, there exists a unique α that rationalizes choices.

[39] To extend to allow for some weak preference, e.g., implications of symmetry, let B be a $D' \times m$ matrix with **columns** given by $p - q$, where p and q represent comparisons including weak preference, and $\vec{0}$ be a $D' \times 1$ matrix of zeros. Repeat the above with $\hat{A} = \begin{bmatrix} A & B \\ \vec{1} & \vec{0} \end{bmatrix}$.

PROOF OF THEOREM 3:

Suppose the assumptions of the theorem are satisfied. In the partial-narrow bracketing model, the marginal utility per dollar of spending on good j in decision k is given by

$$\frac{1}{p_j^{t,k}}\left[(1-\alpha)\frac{\partial u\left(x^{t,1}+x^{t,2}\right)}{\partial x_j}+\alpha\frac{\partial u\left(x^{t,k}\right)}{\partial x_j}\right]$$

for $k=1,2$. By the limit conditions, if $\alpha>0$, then $x_j^{t,k}>0$ for $j=1,2$ and $k=1,2$; thus, $x_1^{t,k}=0$ or $x_2^{t,k}=0$ implies $\alpha=0$. If $\frac{1}{p_2^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_2 = \frac{1}{p_1^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_1$ for both k, then set $\alpha=1$. Now suppose that $\frac{1}{p_2^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_2 \neq \frac{1}{p_1^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_1$ and an interior allocation is chosen in budget set k. The first-order condition for an interior maximizer equates the marginal utility per dollar across the two goods; rearranging this condition, we obtain

$$\alpha=\frac{\frac{1}{p_1^{t,k}}\partial u\left(x^{t,1}+x^{t,2}\right)/\partial x_1-\frac{1}{p_2^{t,k}}\partial u\left(x^{t,1}+x^{t,2}\right)/\partial x_2}{\frac{1}{p_1^{t,k}}\partial u\left(x^{t,1}+x^{t,2}\right)/\partial x_1-\frac{1}{p_2^{t,k}}\partial u\left(x^{t,1}+x^{t,2}\right)/\partial x_2+\frac{1}{p_2^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_2-\frac{1}{p_1^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_1}$$

for $k=1,2$. It remains to verify that the expression on the right-hand side always lies in the interval $(0,1)$. Since $\frac{1}{p_2^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_2 \neq \frac{1}{p_1^{t,k}}\partial u\left(x^{t,k}\right)/\partial x_1$, the solution satisfies $(1-\alpha)\left[\frac{1}{p_1^{t,k}}\frac{\partial u\left(x^{t,1}+x^{t,2}\right)}{\partial x_1}-\frac{1}{p_2^{t,k}}\frac{\partial u\left(x^{t,1}+x^{t,2}\right)}{\partial x_2}\right] = \alpha\left[\frac{1}{p_2^{t,k}}\frac{\partial u\left(x^{t,k}\right)}{\partial x_2}-\frac{1}{p_1^{t,k}}\frac{\partial u\left(x^{t,k}\right)}{\partial x_1}\right]$. Thus, if $\alpha\in(0,1)$, the denominator and numerator have the same sign, and the denominator is strictly larger in absolute value—therefore, the expression is well defined, and thus, α is identified from choices.∎

Joint Implications of Narrow and Broad Bracketing.—To study the joint implications of broad and narrow bracketing, we consider endowment bracketing in a rich domain. We thus suppose that we observe an infinite dataset on all decisions that each consist of a binary part and a second degenerate part summarized by its only available bundle z. Suppose that when a person has bundle $z\in\mathbb{R}_+^n$ in their second part and must make a binary choice, they apply a complete, transitive, continuous, and strictly increasing binary relation $\succsim_z$ (call such a binary relation a *well-behaved preference*). In this domain, we say that a person *broadly brackets* if $x \succsim_z y \Leftrightarrow x+z \succsim_0 y+z$; a person *narrowly brackets* if $x \succsim_z y \Leftrightarrow x \succsim_{z'} y$ for all $z'\in\mathbb{R}_+^n$.

THEOREM 4: *The family of well-behaved preferences* $\{\succsim_z\}_{z\in\mathbb{R}_+^n}$ *satisfies both narrow and broad bracketing if and only if there exists a vector* $u\in\mathbb{R}_{++}^n$ *such that* $x \succsim_z y$ *if and only if* $x\cdot u \geq y\cdot u$.

PROOF:

The "if" direction is immediate. So now consider the "only if" direction. We do so by constructing an additive utility representation for $\succsim_0$. Since $\succsim_0$ is a well-behaved preference relation, for each $x \in \mathbb{R}^n_+$, there exists a unique number $\kappa \in \mathbb{R}_+$ such that $x \sim_0 \kappa e$, where $e = (1, 1, \ldots, 1)$; define $u(x) = \kappa$ for each x and the corresponding κ. By strict monotonicity, u represents $\succsim_0$, and by continuity, u is continuous. Pick any $x, y \in \mathbb{R}^n_+$. By NB, $x \sim_y u(x)e$ and $y \sim_{u(x)e} u(y)e$, so by BB and transitivity, $x + y \sim_0 y + u(x)e \sim_0 u(x)e + u(y)e$, so $u(x + y) = u(x) + u(y)$. Since x, y were arbitrary, $u(x) = u \cdot x$ for some $u \in \mathbb{R}^n_{++}$ by Theorem 1 in Aczél (1966, chap. 5). ∎

Derivations of Predictions and Their Experimental Implementations.—First, observe that if relation P is acyclic, it must be antisymmetric.

Derivation of Prediction 1 (NB-WARP)**:** Suppose $x^{t',k'} \in B^{t,k} \subseteq B^{t',k'}$. By the definition of P in NB-SARP, $x^{t',k'} P x$ for all $x \in B^{t',k'} \backslash \{x^{t',k'}\}$. Since $B^{t,k} \subseteq B^{t',k'}$, it follows that $x^{t',k'} P x$ for all $x \in B^{t,k} \backslash \{x^{t',k'}\}$. But if $x^{t,k} \neq x^{t',k'}$, the definition of P would imply $x^{t,k} P x^{t',k'}$, which would violate acyclicity. Thus, $x^{t,k} = x^{t',k'}$.

Discretized Implementation in Our Experiment.—In all of our tests of NB-WARP, we have $B^{t,k} = B^{t',k'}$ exactly. We thus test NB-WARP exactly, without needing to adjust for discreteness.

Derivation of Prediction 2 (BB-WARP)**:** Suppose $x^{t'} \in B^t \subseteq B^{t'}$. By the definition of P in BB-SARP, $x^{t',k'} P x$ for all $x \in B^{t'} \backslash \{x^{t'}\}$. Since $B^t \subseteq B^{t'}$, it follows that $x^{t'} P x$ for all $x \in B^t \backslash \{x^{t'}\}$. But if $x^t \neq x^{t'}$, the definition of P would imply $x^t P x^{t'}$, which would violate acyclicity. Thus, $x^t = x^{t'}$.

Discretized Implementation in Our Experiment.—In the Risk and Social Experiments, $\mathrm{co}\, B^{1,1} + \mathrm{co}\, B^{1,2} \subseteq \mathrm{co}\, B^{2,1}$. In particular, $B^{2,1} = \{(\$0, \$28); (\$2, \$26); \ldots; (\$28, \$0)\}$, whereas $B^{1,1} + B^{1,2} = \{(\$0, \$28); (\$1, \$27); \ldots; (\$16, \$12); (\$17, \$10.80); \ldots; (\$26, \$0)\}$. Consider two cases.

If $x_A^{2,1} \leq \$16$, the bundle chosen in $B^{2,1}$ is exactly affordable in $B^{1,1} + B^{1,2}$. The caveat is that the set $B^{1,1} + B^{1,2}$ has a higher resolution in this range. We do not adjust for this, implicitly interpreting their choice from $B^{2,1}$ as revealing their preferences over the $\mathrm{co}\, B^{2,1}$. However, if this leads a subject to fail BB-WARP, if their preferences are well behaved, they would choose in $B^{2,1}$ one of the closest bundles to their preferred bundle in $\mathrm{co}\, B^{2,1}$ and be within one error of passing BB-WARP.

However, if $x_A^{2,1} > \$16$, then the person reveals a sufficiently strong preference for person/state A over B that their desired bundle in B^2 is not affordable in co $B^{1,1} + \mathrm{co}\, B^{1,2}$, and thus, they trivially pass BB-WARP.

Derivation of Prediction 3 (BB-Mon)**:** Suppose $x_1^{t,2} > 0$, $x_2^{t,1} > 0$, and $p_1^{t,1} \leq p_2^{t,1}$, $p_1^{t,2} > p_2^{t,2}$. Let $\epsilon = \min\{x_1^{t,2}, x_2^{t,1}\}$. Consider the alternative pair of choices $\left(y^{t,1}, y^{t,2}\right)$ given by $y^{t,1} = \left(x_1^{t,1} + \frac{p_2^{t,1}}{p_1^{t,1}}\epsilon, x_2^{t,1} - \epsilon\right)$ and $y^{t,2} = \left(x_1^{t,2} - \epsilon, x_2^{t,2} + \frac{p_1^{t,2}}{p_2^{t,2}}\epsilon\right)$. By construction, $\left(y^{t,1}, y^{t,2}\right)$ is affordable. We have that

$$\begin{aligned} y_1^{t,1} + y_1^{t,2} &= x_1^{t,1} + x_1^{t,2} + \frac{p_2^{t,1} - p_1^{t,1}}{p_1^{t,1}}\epsilon \geq x_1^{t,1} + x_1^{t,2}, \quad \text{and} \\ y_2^{t,1} + y_2^{t,2} &= x_2^{t,1} + x_2^{t,2} + \frac{p_1^{t,2} - p_2^{t,2}}{p_2^{t,1}}\epsilon > x_2^{t,1} + x_2^{t,2}, \end{aligned}$$

where the inequalities respectively follow from $p_1^{t,1} \leq p_2^{t,1}$ and $p_1^{t,2} > p_2^{t,2}$. But $x^t P y^t$ would violate monotonicity; thus, such an $x^{t,1}, x^{t,2}$ pair could not pass BB-SARP.

The argument in the proof applies with minimal modification to our discretized experiment, and thus, we apply the conditions directly.

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